PROJECT REPORT

Enterprise Network Design with VLANs, Trunking,
Inter-VLAN Routing, DHCP and DNS

Submitted By

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1. Introduction

This project involves the design and configuration of an enterprise network consisting of two routers and four switches. The network implements Virtual Local Area Networks (VLANs) spanning multiple switches using IEEE 802.1Q trunking. Inter-VLAN routing is achieved using the Router-on-a-Stick method on Router0. A DHCP server provides automatic IP addressing for all end devices, and a DNS server handles internal hostname resolution.

2. Objectives

- Design a network using 2 routers (Cisco 1941) and 4 switches (Cisco 2960-24TT)
- Configure VLANs spanning multiple switches using trunk links
- Implement inter-VLAN routing using Router-on-a-Stick on Router0
- Deploy a DHCP server for automatic IP address assignment
- Configure a DNS server for internal hostname resolution
- Verify configuration using show vlan brief and show interfaces trunk commands

3. Network Topology

The network follows a hierarchical design with Router0 as the core routing device, Switch1 as the central distribution switch, and Switch0, Switch2, and Switch3 as access switches serving different departments.

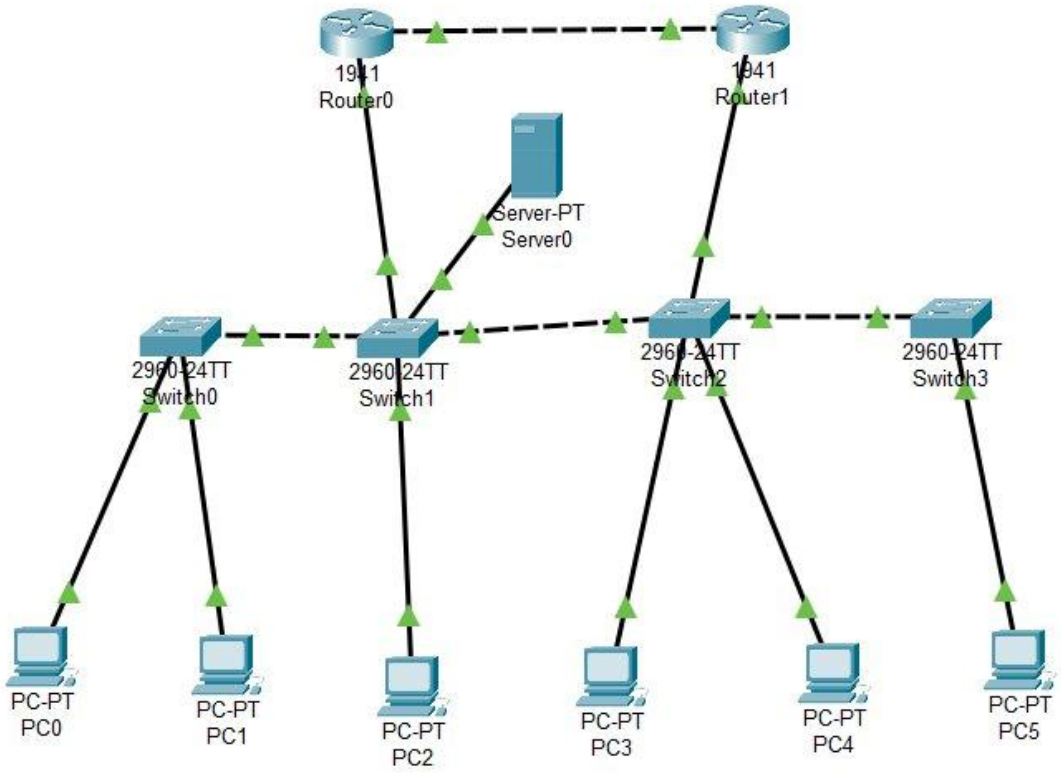

Figure 1: Network Topology Diagram

4. Device & IP Address Plan

4.1 Device Summary

Device	Model	Role	Interface	IP Address
Router0	Cisco 1941	Edge + Inter-VLAN	Gi0/1	10.0.0.1/30
Router1	Cisco 1941	Distribution	Gi0/0	192.168.2.1/24
Router1	Cisco 1941	Distribution	Gi0/1	10.0.0.2/30
Switch1	2960-24TT	Core L2 Switch	VLAN 99	192.168.99.1/24
Switch0	2960-24TT	Dept A Access	VLAN 99	192.168.99.11/24
Switch2	2960-24TT	Dept B/C Dist.	VLAN 99	192.168.99.12/24
Switch3	2960-24TT	Dept C Access	VLAN 99	192.168.99.13/24
Server0	Server-PT	DHCP + DNS	Fa0	192.168.100.200/24

4.2 VLAN Plan

VLAN ID	Name	Subnet	Gateway	Devices
10	DEPT_A	192.168.10.0/24	192.168.10.1	PC0, PC1
20	DEPT_B	192.168.20.0/24	192.168.20.1	PC2, PC3
30	DEPT_C	192.168.30.0/24	192.168.30.1	PC4, PC5
99	MANAGEMENT	192.168.99.0/24	192.168.99.1	All switches
100	SERVERS	192.168.100.0/24	192.168.100.1	Server0

5. VLAN Configuration

VLANs were created on all switches using the following commands (shown for Switch1; repeated on Switch0, Switch2, and Switch3 for their respective VLANs):

```
Switch1# configure terminal
Switch1(config)# vlan 10
Switch1(config-vlan)# name DEPT_A
Switch1(config-vlan)# vlan 20
Switch1(config-vlan)# name DEPT_B
Switch1(config-vlan)# vlan 30
Switch1(config-vlan)# name DEPT_C
Switch1(config-vlan)# vlan 99
Switch1(config-vlan)# name MANAGEMENT
Switch1(config-vlan)# vlan 100
Switch1(config-vlan)# name SERVERS
Switch1(config-vlan)# exit
```

6. Trunk Configuration

Trunk links carry traffic from multiple VLANs between switches using IEEE 802.1Q encapsulation. VLAN 99 is set as the native VLAN on all trunk ports.

6.1 Switch1 Trunk Ports

```
interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,99,100
 switchport trunk native vlan 99
interface FastEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,99
 switchport trunk native vlan 99
interface FastEthernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 20,30,99
 switchport trunk native vlan 99
```

6.2 Access Switch Trunk Uplinks

```
! Switch0 uplink to Switch1
interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,99
! Switch2 uplink to Switch1
interface FastEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 20,30,99
! Switch3 uplink to Switch2
interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 30,99
```

7. Inter-VLAN Routing

Since all switches are Cisco 2960 (Layer 2 only), inter-VLAN routing is implemented using the Router-on-a-Stick method on Router0. Subinterfaces are created on GigabitEthernet0/0, one per VLAN, each with IEEE 802.1Q encapsulation.

```
interface GigabitEthernet0/0
 no ip address
 no shutdown
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.100.200
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.100.200
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 ip helper-address 192.168.100.200
!
interface GigabitEthernet0/0.99
 encapsulation dot1Q 99 native
 ip address 192.168.99.1 255.255.255.0
!
interface GigabitEthernet0/0.100
 encapsulation dot1Q 100
 ip address 192.168.100.1 255.255.255.0
```

8. DHCP Configuration

The DHCP server (Server0 at 192.168.100.200) provides automatic IP addressing for all three department VLANs. The ip helper-address command on each Router0 subinterface relays DHCP broadcasts from clients to the server.

Pool Name	Start IP	Subnet Mask	Gateway	DNS
POOL_VLAN10	192.168.10.11	255.255.255.0	192.168.10.1	.100.200
POOL_VLAN20	192.168.20.11	255.255.255.0	192.168.20.1	.100.200
POOL_VLAN30	192.168.30.11	255.255.255.0	192.168.30.1	.100.200

9. DNS Configuration

The DNS service runs on Server0 alongside DHCP. The following A records were configured through the Packet Tracer Services GUI:

Hostname	Type	IP Address
server0.local	A Record	192.168.100.200
router0.local	A Record	192.168.10.1
switch1.local	A Record	192.168.99.1

10. Verification

10.1 show vlan brief — Switch1

```
Switch1# show vlan brief

VLAN  Name                   Status    Ports
----  --------------------   --------  --------------------
1     default                active
10    DEPT_A                 active
20    DEPT_B                 active
30    DEPT_C                 active
99    MANAGEMENT             active
100   SERVERS                active    Fa0/24
```

10.2 show interfaces trunk — Switch1

```
Switch1# show interfaces trunk

Port     Mode    Encapsulation  Status     Native vlan
Gi0/1    on      802.1q         trunking   99
Fa0/1    on      802.1q         trunking   99
Fa0/2    on      802.1q         trunking   99

Port     Vlans allowed on trunk
Gi0/1    10,20,30,99,100
Fa0/1    10,99
Fa0/2    20,30,99
```

10.3 Connectivity Test Results

Test	Source	Destination	Result	Type
Intra-VLAN ping	PC0 (VLAN 10)	PC1 (VLAN 10)	Pass	Same VLAN
Inter-VLAN ping	PC0 (VLAN 10)	PC2 (VLAN 20)	Pass	Via RoaS
DHCP lease	PC4 (VLAN 30)	Server0	Pass	Auto IP
DNS resolution	PC0	server0.local	Pass	DNS
Server ping	PC0	192.168.100.200	Pass	Cross-VLAN

11. Conclusion

- A two-router, four-switch enterprise network was successfully designed and implemented in Cisco Packet Tracer
- Five VLANs (10, 20, 30, 99, 100) were created and verified across all switches
- IEEE 802.1Q trunk links established between Switch1 and all connected switches
- Inter-VLAN routing implemented using Router-on-a-Stick with subinterfaces on Router0 (Gi0/0.10, .20, .30, .99, .100)
- DHCP relay configured using ip helper-address on each subinterface; all PCs received IPs automatically
- DNS server configured with A records for internal hostname resolution
- Configuration verified using show vlan brief, show interfaces trunk, and ping tests — all passed